Reg. No.										

MANIPAL INSTITUTE OF TECHNOLOGY

MANIPAL

(A constituent unit of MAHE, Manipal)

V SEMESTER B. TECH (MECHANICAL AND INDUSTRIAL AND PRODUCTION ENGINEERING) END SEMESTER EXAMINATIONS, NOVEMBER 2019

SUBJECT: WORK SYSTEMS ENGINEERING [MME 4038]

REVISED CREDIT SYSTEM

Time: 3 Hours MAX. MARKS: 50

Instructions to Candidates:

- Answer **ALL** the questions.
- Missing data may be suitably assumed.

1A. Define Method study. List the objectives of Method study. **(02)**

1B. What is Rating factor? Explain any two rating methods. **(03)**

1C. Define the Basic work content. Explain the management techniques for reducing Excess work content and Ineffective time. **(05)**

2A. Explain the Cumulative and Differential methods of timing the elements in Time study. **(02)**

2B. List the principles of motion economy with regard to design of tools and equipment and arrangement of the workplace. **(03)**

2C. The following observations were made in a Method study on an operator in charge of two identical machines X and Y:

Description of events	Time (min)
Cleaning and checking a finished job.	2
Preparing a job for machining.	4
Stopping and unloading a machine.	3
Loading and starting a machine.	5
Automatic processing by a machine.	22

Draw a Man-machine chart for the best sequence. Find the percent utilization for each resource. What will be the cost per piece if operator costs Rs 60/- per hour and each machine costs Rs 80/- per hour respectively? **(05)**

3A. Explain the effects of noise and vibration with regard to working environment on productivity. **(02)**

3B. With a neat sketch explain the Outline process chart. **(04)**

3C. Eight observations taken for an element in a Time study are as follows. Find out whether the number of observations are sufficient considering 95% confidence level and ± 5% accuracy.

Time in decimal min.
8
8
8
7
7
8
7
8

(04)

4A. Explain the Questioning technique employed in the critical examination phase of Method study. **(03)**

4B. Define Work study. Explain the steps involved in performing Work study. **(03)**

4C. Distinguish between Flow diagram and String diagram. **(04)**

5A. Explain Multiple machine work and Interference allowance. **(02)**

5B. Define Work sampling. Explain the procedure for Work sampling. **(03)**

5C. Calculate the standard time from the data given below with regard to a restricted work and represent the various components in a Pump diagram.

Elements	Average observed time (in decimal units)	Rating (%)
Element A (outside work)	160	70
Element B (outside work)	90	130
Element C (inside work)	120	80

Machine controlled time = 550 decimal units.
P.N.A = 15 %, F.A. = 5 %
(1 min = 100 decimal units) **(05)**